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Three-dimensional memory string array of thin-film ferroelectric transistors formed with an oxide semiconductor channel

Abstract

A memory structure includes storage transistors organized as horizontal NOR memory strings where the storage transistors are thin-film ferroelectric field-effect transistors (FeFETs) having a ferroelectric gate dielectric layer formed adjacent an oxide semiconductor channel region. The ferroelectric storage transistors thus formed are junctionless transistors having no p/n junction in the channel. In some embodiments, the ferroelectric storage transistors in each NOR memory string share a common source line and a common bit line, the common source line and the common bit line formed on a first side of the channel region and the ferroelectric gate dielectric layer and in electrical contact with the oxide semiconductor channel region. The ferroelectric storage transistors in a NOR memory string are controlled by individual control gate electrodes formed on a second side, opposite the first side, of the ferroelectric gate dielectric layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/243,850, entitled THREE-DIMENSIONAL MEMORY STRING ARRAY OF THIN-FILM FERROELECTRIC TRANSISTORS FORMED WITH OXIDE SEMICONDUCTOR CHANNEL, filed Sep. 14, 2021, which is incorporated herein by reference in its entirety for all purposes. (2) This application relates to (i) U.S. non-provisional patent application (“patent application I”), Ser. No. 17/674,137, entitled “Thin-Film Storage Transistor With Ferroelectric Storage Layer,” filed Feb. 17, 2022, and (ii) U.S. non-provisional application (“patent application II”), Ser. No. 17/812,375, entitled “3-Dimensional Memory String Array Of Thin-Film Ferroelectric Transistors,” filed on Jul. 13, 2022. (3) The present application is also related to (i) U.S. non-provisional application (“patent application III”), Ser. No. 17/382,064, entitled “Methods For Fabricating A 3-Dimensional Memory Structure Of NOR Memory Strings,” filed on Jul. 21, 2021; and (ii) U.S. non-provisional application (“patent application IV”), Ser. No. 17/382,126, entitled “Methods for Fabricating A 3-Dimensional Memory Structure of NOR Memory Strings,” filed on Jul. 21, 2021. (4) The present application is also related to U.S. non-provisional patent application (“patent application V”), Ser. No. 16/894,596, entitled “Capacitive-Coupled Non-Volatile Thin-Film Transistor Strings in Three Dimensional Arrays,” filed on Jun. 5, 2020, which is a continuation of U.S. patent application Ser. No. 16/107,118, entitled “Capacitive-Coupled Non-Volatile Thin-Film Transistor Strings in Three Dimensional Arrays,” filed on Aug. 21, 2018, which is a divisional application of U.S. non-provisional patent application Ser. No. 15/248,420, entitled “Capacitive-Coupled Non-Volatile Thin-Film Transistor Strings in Three Dimensional Arrays,” filed on Aug. 26, 2016, now U.S. Pat. No. 10,121,553, issued Nov. 6, 2018, which is related to and claims priority of (i) U.S. provisional application, Ser. No. 62/235,322, entitled “Multi-gate NOR Flash Thin-film Transistor Strings Arranged in Stacked Horizontal Active Strips With Vertical Control Gates,” filed on Sep. 30, 2015; (ii) U.S. provisional patent application, Ser. No. 62/260,137, entitled “Three-dimensional Vertical NOR Flash Thin-film Transistor Strings,” filed on Nov. 25, 2015; (iii) U.S. non-provisional patent application Ser. No. 15/220,375, “Multi-Gate NOR Flash Thin-film Transistor Strings Arranged in Stacked Horizontal Active Strips With Vertical Control Gates,” filed on Jul. 26, 2016; and (vi) U.S. provisional patent application, Ser. No. 62/363,189, entitled “Capacitive Coupled Non-Volatile Thin-film Transistor Strings,” filed Jul. 15, 2016. (5) The disclosures of patent applications I, II, III, IV, and V are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates to high-density memory structures. In particular, the present invention relates to high-density, low read-latency memory structures formed by interconnected thin-film storage elements (e.g., 3-dimensional array of thin-film storage transistors), including those organized as NOR-type memory strings (“NOR memory strings”).

BACKGROUND OF THE INVENTION

(2) A NOR-type memory string includes storage transistors that share a common source region and a common drain region, while allowing each storage transistor to be individually addressed and accessed. U.S. Pat. No. 10,121,553 (the '553 patent), entitled “Capacitive-Coupled Non-Volatile Thin-film Transistor NOR Strings in Three-Dimensional Arrays,” issued on Nov. 6, 2018, discloses storage or memory transistors organized as 3-dimensional arrays of NOR memory strings formed above a planar surface of a semiconductor substrate. The '553 patent is hereby incorporated by reference in its entirety for all purposes. In the '553 patent, a NOR memory string includes numerous thin-film storage transistors that share a common bit line and a common source line. In particular, the '553 patent discloses a NOR memory string that includes (i) a common source region and a common drain region both running lengthwise along a horizontal direction and (ii) gate electrodes for the storage transistors each running along a vertical direction. In the present description, the term “vertical” refers to the direction normal to the surface of a semiconductor substrate, and the term “horizontal” refers to any direction that is parallel to the surface of that semiconductor substrate. In a 3-dimensional array, the NOR memory strings are provided on multiple planes (e.g., 8 or 16 planes) above the semiconductor substrate, with the NOR memory strings on each plane arranged in rows. For a charge-trap type storage transistor, data is stored in a charge storage film in each storage transistor. For examples, the charge storage film includes a tunneling dielectric layer, a charge trapping layer and a blocking layer, which can be implemented as a multilayer including silicon oxide, silicon nitride, and silicon oxide, arranged in this order and referred to as an ONO layer. An applied electrical field across the charge storage film adds or removes charge from charge traps in the charge trapping layer, altering the threshold voltage of the storage transistor to encode a given logical state to the storage transistor.

(3) Advances in electrically polarizable materials (“ferroelectric materials”), especially those that are being used in semiconductor manufacturing processes, suggest new potential applications in ferroelectric memory circuits. For example, the article “Ferroelectricity in Hafnium Oxide: CMOS compatible Ferroelectric Field Effect Transistors,” by T. S. Boscke et al., published in 2011 International Electron Devices Meeting (IEDM), pp. 24.5.1-24.5.4, discloses a ferroelectric field effect transistor (“FeFET”) that uses hafnium oxide as a gate dielectric material. By controlling the polarization direction in a ferroelectric gate dielectric layer, the FeFET may be programmed to have either one of two threshold voltages. Each threshold voltage of the FeFET constitutes a state, for example, a “programmed” state or an “erased” state, that represents a designated logical value. Such an FeFET has application in high-density memory circuits. For example, U.S. patent application Ser. No. 13/897,037, entitled “Apparatuses having a ferroelectric field-effect transistor memory array and related method,” by D. V. Nirmal Ramaswamy et al., filed on May 17, 2013, now U.S. Pat. No. 9,281,044, discloses a 3-dimensional array of FeFETs.

(4) The FeFETs of the prior art, however, suffer from low endurance. For example, the article “Vertical Ferroelectric HfO₂ FET based on 3-D NAND Architecture: Towards Dense Low-Power Memory,” by K. Florent et al., published in 2018 IEEE International Electron Devices Meeting (IEDM), 2018, pp. 2.5.1-2.5.4, discloses an endurance of merely 10⁴ cycles. Such low endurance renders the memory circuits practically unsuitable for many memory applications.

SUMMARY OF THE INVENTION

(5) In embodiments of the present invention, a three dimensional memory structure formed above a planar surface of a semiconductor substrate includes a set of thin-film ferroelectric field-effect

transistors (FeFETs) formed above the semiconductor substrate and arranged in one or more parallel planes, wherein (i) the thin-film FeFETs are organized as NOR memory strings where each NOR memory string extends along a first direction substantially parallel to the planar surface of the semiconductor substrate; (ii) the NOR memory strings are arranged in two directions: (a) as stacks of NOR memory strings where each stack including NOR memory strings is arranged one on top of another along a second direction substantially normal to the planar surface and the NOR memory strings within each stack are isolated from another by a first isolation layer separating adjacent planes, and (b) as rows of NOR memory strings arranged along a third dimension substantially orthogonal to both the first dimension and the second direction; (iii) the FeFETs within each NOR memory string share a common source line and a common bit line where the common source line and the common bit line each extends along the first direction and arranged spaced apart from each other in the second direction by a second isolation layer; (iv) each FeFET in each NOR memory string includes an oxide semiconductor layer and a ferroelectric gate dielectric layer formed adjacent the oxide semiconductor layer where the oxide semiconductor layer forms a channel region between the common source line and the common bit line and the oxide semiconductor layer is separated by the first isolation layer to each NOR memory string within each stack; and (v) a set of conductors each extends along the second direction where each conductor is provided between adjacent stacks of NOR memory strings and serves as a common gate electrode for respective FeFETs in the NOR memory strings of the adjacent stacks.

(6) In some embodiments, the ferroelectric gate dielectric layer includes a ferroelectric polarization layer provided adjacent each conductor as a continuous layer in the second direction.

(7) In some embodiments, the ferroelectric gate dielectric layer is formed of a doped hafnium oxide material and the oxide semiconductor layer forming the channel region is an amorphous oxide semiconductor material.

(8) In some embodiments, the common bit line and the common source line of each NOR memory string are made partially or substantially of a metallic conductor material.

(9) These and other advantages, aspects and novel features of the present invention, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments of the invention are disclosed in the following detailed description and the accompanying drawings. Although the drawings depict various examples of the invention, the invention is not limited by the depicted examples. It is to be understood that, in the drawings, like reference numerals designate like structural elements. Also, it is understood that the depictions in the figures are not necessarily to scale.

(2) FIG. 1 is a perspective view of a memory structure including a 3-dimensional array of NOR memory strings in embodiments of the present invention.

(3) FIG. 2 illustrates the detail construction of a storage transistor that can be formed in the memory structure of FIG. 1 in some embodiments.

(4) FIG. 3(a) illustrates the detail construction of a storage transistor with separated ferroelectric gate dielectric layer in some embodiments.

(5) FIG. 3(b) illustrates the detail construction of a storage transistor with separated ferroelectric gate dielectric layer in alternate embodiments.

(6) FIG. 4 is a circuit representation of a 3-dimensional array of NOR memory strings in embodiments of the present invention.

(7) FIGS. 5(a) and 5(b) illustrate the erase and program operations of the ferroelectric storage

transistor in embodiments of the present disclosure.

(8) FIGS. 6(a) to 6(q) illustrate a process for fabricating a memory structure including a 3-dimensional array of NOR memory strings of ferroelectric storage transistors according to embodiments of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(9) According to embodiments of the present invention, a memory structure includes storage transistors organized as horizontal NOR memory strings where the storage transistors are thin-film ferroelectric field-effect transistors (FeFETs) having a ferroelectric gate dielectric layer formed adjacent an oxide semiconductor channel region. The ferroelectric storage transistors thus formed are junctionless transistors having no p/n junction in the channel and which the mobile carrier density in the channel is modulated by the polarization of the ferroelectric gate dielectric layer. In embodiments of the present invention, the ferroelectric storage transistors in each NOR memory string share a first conductive layer functioning as a common source line and a second conductive layer functioning as a common bit line, the first and second conductive layers being in electrical contact with the oxide semiconductor channel region. The ferroelectric storage transistors in a NOR memory string are controlled by individual control gate electrodes to allow each storage transistor to be individually addressed and accessed. In some embodiments, the ferroelectric gate dielectric layer is formed of a doped hafnium oxide material and the oxide semiconductor channel region is formed of an amorphous oxide semiconductor material.

(10) In some embodiments, the ferroelectric storage transistors in each NOR memory string share a common source line and a common bit line that are formed on a first side of the channel region, away from the ferroelectric gate dielectric layer, and in electrical contact with the oxide semiconductor channel region. The ferroelectric storage transistors in a NOR memory string are controlled by individual control gate electrodes that are formed adjacent the ferroelectric gate dielectric layer on a second side, opposite the first side, of the channel region.

(11) In some embodiments, the memory structure includes multiple NOR memory strings organized in a three dimensional array to form a high density memory structure. The three dimensional array of NOR memory strings is organized as stacks of NOR memory strings in a first direction, with NOR memory strings formed one on top of another in a stack in the first direction. The three dimensional array of NOR memory strings is also organized as rows of NOR memory strings in a second direction forming a plane, with the rows of NOR memory strings arranged in one or more parallel planes in first direction. In some embodiments, the shared common source line is electrically floating and the source voltage is applied from the common bit line using pre-charge transistors so as to mitigate congestion of connector wires at the bit-line/source-line end staircases in the three-dimensional structure.

(12) The ferroelectric storage transistors, as described herein, provide high endurance, long data retention, and relatively low voltage operations for both erase (under 5.0 volts) and programming (e.g., under -5.0 volts) operations. By combining the ferroelectric or polarization characteristics with the 3-dimensional organization (e.g., as the thin-film NOR memory strings described herein), the memory structure of ferroelectric storage transistors of the present invention achieves the additional benefits of high-density, low-cost memory arrays with the advantages of high-speed, randomly accessed memory circuits with low read latency. These and other advantages of the memory structure of the present invention will be described further in the following description.

(13) In the present description, the term “storage transistor” is used interchangeably with “memory transistor” to refer to the memory device formed in the memory structure described herein. In some examples, the memory structure of the present disclosure including NOR memory strings of randomly accessible storage transistors (or memory transistors) can have applications in computing systems as the main memory where the memory locations are directly accessible by the processors of the computer systems, for instance, in a role served in the prior art by conventional random-access memories (RAMs), such as dynamic RAMs (DRAMs) and static RAMs (SRAMs). For

example, the memory structure of the present disclosure can be applied in computing systems to function as a random-access memory to support the operations of microprocessors, graphical processors and artificial intelligence processors. In other examples, the memory structure of the present disclosure is also applicable to form a storage system, such as a solid-state drive or replacing a hard drive, for providing long term data storage in computing systems.

(14) In the present description, to facilitate reference to the figures, a Cartesian coordinate reference frame is used, in which the Z-direction is normal to the planar surface of the semiconductor surface and the X-direction and the Y-directions are orthogonal to the Z-direction and to each other, as indicated in the figures.

(15) Furthermore, the drawings provided herein are idealized representations to illustrate embodiments of the present disclosure and are not meant to be actual views of any particular component, structure, or device. The drawings are not to scale, and the thickness and dimensions of some layers may be exaggerated for clarity. Variations from the shapes of the illustrations are to be expected. For example, a region illustrated as a box shape may typically have rough and/or nonlinear features. Sharp angles that are illustrated may be rounded. Like numerals refer to like components throughout.

(16) Memory Structure

(17) FIG. 1 is a perspective view of a memory structure including a 3-dimensional array of NOR memory strings in embodiments of the present invention. The memory structure can be used to implement part of a semiconductor memory device in some examples. Referring to FIG. 1, a memory structure **10** includes a number of alternating conductive and isolation layers formed on a planar surface of a semiconductor substrate **12**. In the present description, a pair of conductive layers with an interleaving isolation layer are referred to as an active layer **11**. A buffer oxide layer **14** may be provided between the semiconductor substrate **12** and the active layers **11** formed on the substrate. The active layers **11**, including alternating conductive and isolation layers, are formed one on top of another in the Z-direction (i.e., along a direction normal to the planar surface of the substrate **12**). The active layers **11** are divided in the X-direction into narrow strips (“active strips”) **24** that are stacked one on top of another to form stacks of active strips (“active stacks”) extending in the Y-direction. As thus formed, each active strip **24** form a NOR memory string of ferroelectric storage transistors **20** extending in the Y-direction.

(18) Each active layer **11** includes a first conductive layer **16** functioning as a common source line for the NOR memory string and a second conductive layer **18** functioning as a common bit line for the NOR memory string. The first and second conductive layers **16**, **18** are separated by an isolation layer **17**. In the present illustration, eight active layers **11-0** to **11-7** are provided. Each active layer is separated from the others by an isolation layer **15**. The isolation layers **15**, **17** can each be an insulating dielectric layer or other suitable isolation layer.

(19) Subsequent processing steps form oxide semiconductor channel regions **25**, ferroelectric gate dielectric layers **26** and gate electrodes **28** in narrow trenches **22** between the separated active stacks. In some embodiments, the oxide semiconductor channel region **25** is isolated to each NOR memory string. In the present embodiment, the oxide semiconductor channel region **25** is separated at each NOR memory string by the isolation layer **15**.

(20) In embodiments of the present invention, the gate electrodes **28** and the ferroelectric gate dielectric layers **26** are formed as columnar structures extending in the Z-direction. In the present example, ferroelectric gate dielectric layer **26** encircles a gate electrode **28** in the columnar structure. In the present description, the gate electrodes **28** are also referred to as “local word lines” and a gate electrode **28** encircled by a ferroelectric gate dielectric layer **26** is collectively referred to a local word line (LWL) structure. Local word line structures formed in each trench **22** are isolated from each other by a dielectric material **23**. A ferroelectric storage transistor **20** is formed at the intersection of an active strip **11** with the channel region **25** and an LWL structure. Accordingly, ferroelectric storage transistors **20** are formed on both sides of the active strip. In the present

illustration, the LWL structures are formed staggered in adjacent trenches **22** so that storage transistors **20** formed on both sides of the active strip are offset from each other in the Y-direction along the NOR memory string. In particular, isolation layer **15** between the NOR memory strings and isolation layer **17** between adjacent source line and bit line provides isolation to decouple the ferroelectric storage transistors formed on the two sides of the same active strip. As thus configured, along each active strip (in the Y-direction), the ferroelectric storage transistors **20** that share the common source line **16** and the common bit line **18** form a NOR memory string (also referred as a “Horizontal NOR memory string” or “HNOR memory string”).

(21) In the 3-dimensional array of NOR memory strings thus formed, the ferroelectric storage transistors **20** are junctionless transistors that do not include p/n junction as the drain or source region in the channel. Instead, the second conductive layer **18** (common bit line) serves as the drain terminal and the first conductor layer **16** serves as the source terminal of the ferroelectric storage transistor **20**. Accordingly, a NOR memory string includes ferroelectric storage transistors **20** that share a common drain terminal (the common bit line **16**) and a common source terminal (the common source line **18**).

(22) In some embodiments, the first and second conductive layers **16** and **18** are each formed using a metal layer or a low resistivity metallic conductive material, such as molybdenum (Mo), tungsten (W), tungsten nitride (WN), ruthenium or titanium tungsten alloy (TiW). In some embodiments, the isolation layer **15** and **17** may be formed as a silicon oxide layer (SiO₂). In other embodiments, the isolation layer **15** may be formed by other isolation dielectric materials, such as silicon nitride. In some embodiments, the ferroelectric gate dielectric layer is formed of a doped hafnium oxide material, such as zirconium-doped hafnium oxide (HfZrO or “HZO”), and the oxide semiconductor channel region is formed of an amorphous oxide semiconductor material, such as indium gallium zinc oxide (IGZO).

(23) To complete the memory circuit, various types of circuits are formed in or at the surface of the semiconductor substrate **12** to support the memory operations of the HNOR memory strings. Such circuits are referred to as “circuit under array” (“CuA”) and may include analog and digital circuits. For example, the memory operations may include erase, program and read operations. In the present description, a write operation to the memory circuit includes the erase and the program operations. Furthermore, in some embodiments, the memory operations include a refresh operation. In some embodiments, the circuit under array supports the memory operations of the memory circuit, including erase, program, read and refresh operations.

(24) In some embodiments, the circuit under array includes various voltage sources or voltage generators for generating operating voltages, such as power supply voltages, ground voltages, programming, erase or read voltages, or reference voltages. The circuit under array may further include word line driver circuits, bit line driver circuits, and input/output driver circuits. The circuit under array may further include address decoders for decoding address signals to select designated storage transistors, sense amplifiers to read stored data from the selected storage transistors, latches and registers, such as shift registers, or other memory elements. The circuit under array may further include various logic circuits, such as inverters, NAND, NOR, Exclusive-Or and other logic gates. In some embodiments, the circuit under array includes state machines, micro-sequencers, and data processing circuitry. For example, in one embodiment, the circuit under array includes a state machine for managing the memory operations (e.g. read, erase, program and refresh operations) at the memory circuit.

(25) In some embodiments, other conductive layers may be arranged above or below the memory structure **10** to provide control signals, such as the word line signals, to the array of ferroelectric storage transistors. In some embodiments, the conductive layers may be arranged to connect the circuit under array to the common bit lines of the NOR memory strings and to the local word lines to support the memory operation. In one embodiment, the conductive layers may be provided for routing control and data signals among the NOR memory strings and the circuit under array. As

thus configured, the circuit under array supports memory operations of the NOR memory strings and implements erase, program and read operations for the NOR memory strings autonomously in response to erase, program and read commands provided to the memory structure. In one example, the write operation to the ferroelectric storage transistors includes an erase operation followed by a program operation.

(26) In some embodiments, the memory structure **10** is coupled to a controller integrated circuit to form a memory module. The controller integrated circuit may be fabricated on a separate semiconductor substrate and electrically connected to the memory structure using one or more integration techniques, including, for example, hybrid bonds, TSVs, exposed contacts and other suitable interconnect techniques. The controller integrated circuit provides commands, such as erase, program and read commands, to the circuit under array, usually with accompanying information, such as the memory cell address and write data for the write operation. The memory structure, using the circuit under array, performs the memory operation autonomously in response to the received command.

(27) In embodiments of the present disclosure, the memory structure **10** represents a modular memory unit, referred to as a “tile,” and a memory device is formed using an array of the modular memory units. In one exemplary embodiment, a memory device is organized as a two-dimensional array of tiles, where each tile includes a three-dimensional array of ferroelectric storage transistors with support circuitry for each tile formed under the respective tile. More specifically, the support circuitry for the ferroelectric storage transistors of each tile is provided for modularity in the portion of the semiconductor substrate underneath each tile. In this manner, each modular memory unit (or tile) operates autonomously to perform its own erase, program, read and refresh operations. As a result, a memory device includes an array of tiles where each tile can be accessed independently of each other with memory operations being performed on multiple tiles concurrently. The memory device may be coupled to a controller integrated circuit to form a memory module. The controller integrated circuit, sometimes referred to as a “chiplet,” implements management functions for the memory device. In some embodiments, the controller integrated circuit provides commands, such as erase, program, read and refresh commands, to the memory device and accompanying command information, such as memory address and write data. The controller integrated circuit may also provide host interface functions, implementing memory interfaces for host access and other system functions. A memory device formed with an array of tiles realizes a high speed and high capacity memory with capabilities for parallel access to storage transistors in the multiple tiles.

(28) FIG. 2 illustrates the detail construction of a storage transistor that can be formed in the memory structure of FIG. 1 in some embodiments. Referring to FIG. 2, a storage transistor **30** is formed by the first conductive layer **16** forming the common source line, the second conductive layer forming the common bit line, and the oxide semiconductor channel region **25** in contact with both the source line **16** and the bit line **18**. The ferroelectric gate dielectric layer **26** is provided between the channel region **25** and the gate electrode **28**. Each storage transistor **30** is isolated from adjacent storage transistors along an active strip (in the Z-direction) by the isolation layer **15**. As thus configured, along each active strip (in the Y-direction), a NOR memory string includes storage transistors **30** formed on both sides of the active strip. In the present embodiment, the local word line structures are staggered in the Y direction on either side of an active strip so that storage transistors **30** are not formed directly across the active strip in the X-direction. In the embodiment shown in FIG. 2, the oxide semiconductor layer are provided only at each local word line structure and is removed between local word line structures which is beneficial to reduce parasitic capacitance. Accordingly, as shown in FIG. 2, the oxide semiconductor channel region **25** is shown only on one side of each active strip. Furthermore, in the present embodiment, the oxide semiconductor channel region **25** is formed in a recess between a pair of isolation layer **15**. Therefore, the channel region **25** is formed within each NOR memory string.

(29) In storage transistor **30** of FIG. **2**, the ferroelectric gate dielectric layer **26** is formed as a continuous layer across each active stack. In other embodiments, the ferroelectric gate dielectric layer **26** may be separated or singulated at each NOR memory string, which can enhance the isolation of the ferroelectric storage transistors in adjacent planes. FIG. **3(a)** illustrates the detail construction of a storage transistor with separated ferroelectric gate dielectric layer in some embodiments. Referring to FIG. **3(a)**, a storage transistor **40** is formed with the ferroelectric gate dielectric layer **26** separated at each NOR memory string. In some embodiments, the ferroelectric gate dielectric layer **26** can be formed using selective deposition so that the ferroelectric gate dielectric layer is only formed adjacent the oxide semiconductor channel region **25** and is not formed on the isolation layer **15**. FIG. **3(b)** illustrates the detail construction of a storage transistor with separated ferroelectric gate dielectric layer in alternate embodiments. Referring to FIG. **3(b)**, a storage transistor **50** is formed with the ferroelectric gate dielectric layer **26** separated at each NOR memory string by the isolation layer **15**. In particular, in some embodiments, the ferroelectric gate dielectric layer **26** can be formed in a recess between a pair of isolation layer **15**. Accordingly, the oxide semiconductor layer and the ferroelectric gate dielectric layer are both formed in the recess to be isolated from other NOR memory strings by the isolation layer **15**.

(30) An oxide semiconductor channel region realizes many advantages in the 3-dimensional array of NOR memory strings of the present invention. First, an oxide semiconductor channel region typically has high mobility for greater switching performance and without concern for electron or hole tunneling. Furthermore, the oxide semiconductor channel has a wider energy bandgap (e.g. 3.1 eV) as compared to a silicon channel (e.g. 1.1 eV). Accordingly, the effect of GIDL (gate induce drain leakage) in the oxide semiconductor channel is much smaller. The storage transistor thus formed has better cell Ion/Ioff ratio (e.g., greater than 1.0×10^7). Good Ion/Ioff ratio is advantageous to the HNOR memory string structure as a large number of storage transistors are connected to the common bit line.

(31) Second, the storage transistors of the NOR memory strings with the oxide semiconductor channel region become junctionless transistors, thereby eliminating the need to provide polysilicon source-drain layers in the memory structure. As a result, the stack height of the 3-dimensional array of NOR memory strings is greatly reduced as compared to traditional memory array with the same number of planes or layers of NOR memory strings. In some embodiments, the oxide semiconductor channel region **25** has a thickness of 8-15 nm in the X-direction. In one example, the oxide semiconductor channel region **25** may have a thickness of 10 nm. In some embodiments, the ferroelectric gate dielectric layer **26** has a thickness of 3-7 nm in the X-direction. In one example, the ferroelectric gate dielectric layer **26** may have a thickness of 4 nm. In some embodiments, the channel length of the ferroelectric storage transistors, that is, the distance between the first and second conductive layers **16**, **18** in the Z-direction, is about 50 nm. In the present description, the dimensions are provided merely for illustrative purposes and are not intended to be limiting. In actual implementation, any suitable thicknesses or dimensions may be used.

(32) Third, the NOR memory strings of junctionless storage transistors formed with the oxide semiconductor channel realize further advantages in processing as well as device reliability. By eliminating the need for polysilicon source-drain layers, high process temperature steps can be eliminated and reliability issues, such as junction breakdown voltage and junction related hot electron degradation, can also be avoided.

(33) In the memory structures described above, the 3-dimensional array of NOR memory strings includes ferroelectric storage transistors that are formed on both side edges along the lengths (Y-direction) of each active strip, with vertical local word lines being provided along both side edges in staggered arrangement in the Y-direction. High density is achieved by sharing the local word lines between adjacent active stacks where each local word line serves as the gate electrode to vertically aligned ferroelectric storage transistors of the active strips of each active stack. The vertical local word lines may be contacted by interconnection conductors (also referred to as

“global word lines”) provided above or below the 3-dimensional array of NOR memory strings. In one embodiment, the global word lines run along a direction (X) that is transverse to the lengths (Y) of the contacted active strips. The global word lines connects the local word lines to the support circuits formed in the circuit under array (CuA) of the semiconductor substrate **12**.

(34) Memory Circuit and Operation

(35) FIG. **4** is a circuit representation of a 3-dimensional array of NOR memory strings in embodiments of the present invention. In particular, FIG. **4** illustrates a portion of the NOR memory strings which can be constructed from the memory structures described above. Referring to FIG. **4**, a memory array **90** includes ferroelectric storage transistors **92** arranged in NOR memory strings **95**, including NOR memory strings **95-0** to **95-3**. Each NOR memory string **95** includes ferroelectric storage transistors **92** that are connected across a common bit line BL **94** (e.g. BL_m to BL_m+3) and a common source line SL **96** (e.g. SL_m to SL_m+3), with each ferroelectric storage transistor **92** in each NOR memory string **95** being controlled by a respective word line WL **98** (e.g. WL_n to WL_n+6). More specifically, each ferroelectric storage transistor **92** in a NOR memory string **95** has a gate terminal connected to the word line WL **98**, a drain terminal connected to the bit line BL **94**, and a source terminal connected to a source line SL **96**.

(36) Memory array **90** includes NOR memory strings formed in an active stack, with each NOR memory strings **95-0** to **95-5** being located on a different plane of the 3-dimensional memory structure. Ferroelectric storage transistors **92** in a column across the multiple NOR memory strings are controlled by a common word line WL **98**, also referred to as a local word line or LWL. FIG. **4** illustrates a portion of the NOR memory strings including ferroelectric storage transistors **92** connected to word lines WL_n to WL_n+6. In practice, there may be thousands of word lines (e.g. **4096** word lines) in the memory array, resulting in thousands of randomly accessible ferroelectric storage transistors in a NOR memory string. The memory array **90** includes a control circuit for controlling the read, write and refresh operations of the memory array. For example, the control circuit can be formed in the semiconductor substrate **12** under the memory array, as shown in FIG. **1**, and is referred to as a circuit under array or CuA. In some examples, the memory array **90** may be organized in memory pages where a memory page refers to a group of storage transistors that are accessed together for read and write operations. The ferroelectric storage transistors **92** in the memory array **90** can be grouped in various ways to form memory pages of the desired size.

(37) In memory array **90**, the source line **96** is electrically floating, that is, the source line **96** is not connected to any electrical potential. In practice, the source line **96** maintains a relatively constant voltage through the parasitic capacitance at the source terminals, such as the parasitic capacitance between the source terminals and the gate terminals of the storage transistors. More specifically, the parasitic capacitance intrinsic to each NOR memory string (e.g., the distributed capacitance between the common source terminal of a NOR memory string and its multiple associated local word lines) may be used as a virtual voltage source, under some operating conditions, to provide a common source voltage. In the present description, the source line **96** is sometimes referred to as being connect to a virtual ground. In other embodiments, the source line **96** can be provided with a given voltage potential, such as the ground voltage, by a voltage source. In some examples, the source voltage can be set to a desired voltage value through a pre-charge operation and the source line can then be left floating after the pre-charge operation. In one embodiment, the pre-charge operation set the bit line to a desired voltage through a pre-charge transistor and then the source line is charged from the bit line to a voltage to equal to the bit line voltage.

(38) In embodiments of the present invention, the ferroelectric storage transistors **92** are thin-film ferroelectric field-effect transistors. More specifically, a ferroelectric field-effect transistor (also referred to as FeFET) is formed by using a ferroelectric material as the gate dielectric layer between the gate conductor, usually a metal layer, and the channel of a field-effect transistor. In some embodiments, the ferroelectric storage transistor may further include an interface dielectric layer adjacent the channel and the ferroelectric gate dielectric layer. The ferroelectric storage

transistor realizes memory function by storing data as polarization states in the ferroelectric gate dielectric layer. In particular, a voltage applied between the gate conductor and the channel induces electrical polarization in the ferroelectric gate dielectric layer, which polarization can be reversed with application of a voltage in the opposite polarity. The ferroelectric gate dielectric layer retains the induced polarization state after removal of the applied gate voltage, realizing the memory function of the ferroelectric storage transistor. For instance, ferroelectric storage transistors can be applied to form non-volatile memory cells where data is stored in the ferroelectric dielectric layer as two stable, remanent polarization states.

(39) The induced polarization states of the ferroelectric gate dielectric layer change the threshold voltage of the ferroelectric storage transistor. The change or shift in the threshold voltage of the ferroelectric storage transistor due to the different polarization states can be used to represent data in different logical states. For example, two logical states (e.g., “0” and “1”) can be represented by the higher and lower threshold voltages of the ferroelectric storage transistor as a result of two induced electrical polarization states in the ferroelectric dielectric layer. In other embodiments, more than two polarization states can be induced in the ferroelectric gate dielectric layer, such as by applying two or more different voltage ranges across the ferroelectric dielectric layer. In this manner, the ferroelectric storage transistor can be operated to store data in more than two logical states.

(40) In embodiments of the present invention, the three-dimensional array of NOR memory strings of ferroelectric storage transistors can be applied to implement a non-volatile memory device or a quasi-volatile memory device. For example, a quasi-volatile memory has an average retention time of greater than 100 ms, such as about 10 minutes or a few hours, whereas a non-volatile memory device may have a minimum data retention time exceeding 5 years. In one embodiment, the ferroelectric storage transistor has a data retention time longer than 1 hour and a program/erase cycle endurance greater than 10^{10} program/erase cycles. For example, the ferroelectric storage transistor may have a data retention time of a few hours and a program/erase cycle endurance of around 10^{11} . Such a ferroelectric storage transistor can be used to form a quasi-volatile memory device where the refresh intervals can be on the order of hours, significantly longer than the refresh intervals of DRAMs which require much more frequent refreshes, such as in tens of milli-seconds. In some embodiments, multiple three-dimensional arrays of ferroelectric storage transistors (“tiles”) may be used to form a quasi-volatile memory device with high memory capacity.

(41) The ferroelectric storage transistor stores data in the ferroelectric gate dielectric layer. In operation, an applied electric field of a first polarity, such as by applying a voltage of the first polarity at the gate terminal relative to the source terminal, induces a first polarization state in the ferroelectric gate dielectric layer. Meanwhile, the application of an electric field of a second polarity, opposite the first polarity, such as by applying a voltage of the second polarity at the gate terminal relative to the source terminal, induces a second polarization state in the ferroelectric gate dielectric layer. The first polarization state shifts the threshold voltage V_{th} of the ferroelectric storage transistor to a lower value, which can be used to encode a first logical state, such as a logical “1” state. Alternately, the second polarization state shifts the threshold voltage V_{th} of the ferroelectric storage transistor to a higher value, which can be used to encode a second logical state, such as a “0” state. In the present description, shifting the threshold voltage V_{th} of the ferroelectric storage transistor to a higher value (logical “0”) is referred herein as a program operation and shifting the threshold voltage V_{th} of the ferroelectric storage transistor to a lower value (logical “1”) is referred herein as an erase operation.

(42) In some embodiments, the pre-charge operation described above can be used to set the source line and bit line voltages of a NOR memory string to voltages that are more negative than the voltage on the local word line (for example during an erase operation), or more positive than the voltage on the local word line (for example during a program operation) are advantageous because

the erase or program polarization states of the ferroelectric storage transistors are somewhat symmetrical in that the cell polarization can be readily reversed by reversing the voltage of the source and drain terminals as one electrode and the voltage of the local word line as the second electrode. In some embodiments, each string of ferroelectric storage transistors, by virtue of being thin-film-transistors, as opposed to a single crystalline transistors formed within the semiconductor substrate, does not require a hard wire connection to the ground potential of the semiconductor substrate for any of its common source, common drain, or common channel. Instead, both polarization states can be imposed on the FeFET with only positive voltages (or only negative voltages) applied to the local word lines or the common source/common drain. This single voltage polarity feature simplifies the process flow for the control/logic transistors formed in the substrate, and allows optimized operation because ferroelectric polarization states typically have a voltage hysteresis that can go from negative voltages (erased state) to positive voltages (programmed state).

(43) In embodiments of the present invention, the ferroelectric storage transistors are formed using an oxide semiconductor channel. For example, the oxide semiconductor channel can be formed using an amorphous oxide semiconductor material, such as indium gallium zinc oxide (InGaZnO or "IGZO"). An oxide semiconductor channel region has the advantage of a high mobility for greater switching performance and without concern for electron or hole tunneling. For example, an IGZO film has an electron mobility of 10.0-100.0 cm²/V, depending on the relative compositions of indium, gallium and zinc. The oxide semiconductor channel forms an N-type, unipolarity channel region where the conductive layers **16a**, **18**, **16b** (FIG. **1**) forming the source and drain terminals directly contact the channel region. The ferroelectric storage transistor thus formed is a depletion mode device where the transistor is normally on and can be turned off by depleting the N-type carriers in the channel region. The threshold voltage of the ferroelectric storage transistor is a function of the thickness (X-direction) of the oxide semiconductor channel region **25**. That is, the threshold voltage of the ferroelectric storage transistor is the amount of voltage necessary to deplete the thickness of the oxide semiconductor channel region to shut off the ferroelectric storage transistor.

(44) Exemplary operation conditions of the three-dimensional NOR memory strings of ferroelectric storage transistors will now be described with reference to FIGS. **5(a)** and **5(b)**. In embodiments of the present invention, the write operation to the NOR memory strings is carried out by first performing an erase operation followed by a program operation. In some cases, the erase operation may erase a memory page of ferroelectric storage transistors and then selective storage transistors are programmed to the desired data value.

(45) As described above, in the three dimensional array of NOR memory string, the ferroelectric storage transistors in a NOR memory string share a common source line and a common bit line. A salient feature of the ferroelectric storage transistors of the present invention is that the common source line (the source terminal) and the common bit line (the drain terminal) are formed on a first side of the channel region and the ferroelectric gate dielectric layer and in electrical contact with the oxide semiconductor channel region. Meanwhile, the ferroelectric storage transistors in the NOR memory string are controlled by individual control gate electrodes which are formed on a second side, opposite the first side, of the ferroelectric gate dielectric layer. This structure is shown in FIGS. **5(a)** and **5(b)**. The configuration of having the drain and source terminals on the opposite side of the channel region/ferroelectric gate dielectric layer as the gate electrode is particularly advantageous for ferroelectric storage transistors to induce the desired polarization state in the ferroelectric gate dielectric layer. In particular, the electric field generated by the voltage difference between the gate electrode and the source/drain terminals can be efficiently applied to induce or change the polarization state of the ferroelectric gate dielectric layer.

(46) More specifically, in the ferroelectric storage transistor of the present invention, the electric field is applied directly across the channel region and the ferroelectric gate dielectric layer to maximize the effect of the electric field for inducing polarization in the ferroelectric gate dielectric

layer. Furthermore, with the source and drain terminals on opposite side of the channel region/ferroelectric gate dielectric layer as the gate electrode, both the electric field and the fringing field from the source and drain terminals to the gate electrode go through the channel region and the ferroelectric gate dielectric layer, maximizing the available electric field to induce the desired polarization. With efficient use of the available electric field, including the fringing field, the ferroelectric storage transistor thus configured can be erased or programmed at a lower erase or program voltages, that is lower voltage difference to be applied across the gate and source/drain terminals. In some cases, ferroelectric memory cell with shorter channel length can realize a wider memory window, or wider threshold voltage difference, due to the higher fringing field penetrating into the channel region. This is contrary to conventional ferroelectric transistor structures where the gate terminal and the source/drain terminals are both formed on the same side of the channel region/ferroelectric gate dielectric layer. In that case, only a small fraction of the electric field and the fringing field from the source and drain terminals to the gate electrodes goes through the channel region and the ferroelectric gate dielectric layer to induce polarization.

(47) Referring to FIG. 5(a), to erase a ferroelectric storage transistor in a selected page, a word line voltage that is more positive than the source line and bit line voltages is used. In some embodiments, the source line voltage and bit line voltage are both set to 0V and the word line or gate voltage is driven to 2-3V. Unselected ferroelectric storage transistors can be inhibited from being erased by having their gate voltage, the bit line voltage and the source voltage to be driven to 1.1V. As a result of the voltage bias between the gate and the drain/source of the transistor, the ferroelectric gate dielectric layer is induced to a first polarization state where electrons are accumulated in the channel region 25 and the threshold voltage of the storage transistor is set to a low voltage value, in particular, lower than the read voltage value. In other words, when the read voltage is applied to the gate 28, the erased ferroelectric storage transistor is conducting. In the present description, ferroelectric storage transistors that have been erased have a first logical state of “1” and ferroelectric storage transistors that have been programmed have a second logical state of “0”.

(48) Referring to FIG. 5(b), to program the ferroelectric storage transistor, a word line voltage that is more negative than the source line and bit line voltages is used. Alternately, the source line voltage and the bit line voltage of the ferroelectric storage transistor to be programmed can be made more positive than the word line voltage. In embodiments of the present disclosure, the program operation of the ferroelectric storage transistors implements asymmetric programming. Asymmetric programming refers to using different source line voltage and the bit line voltage for the program operation. In the present embodiment, to program the selected ferroelectric storage transistor, the word line or gate voltage is set to 0V. The bit line voltage for the selected ferroelectric storage transistor to be programmed is set to 1.8V and the source line voltage is set to 0.8V. To inhibit programming of the unselected ferroelectric storage transistors, the gate voltage, the bit line and source line voltage are all set to 0.8V. As a result of the voltage bias between the gate and the drain/source of the transistor, the ferroelectric gate dielectric layer is induced to a second polarization state where electrons are depleted in the channel region 25 and the threshold voltage of the storage transistor is set to a higher voltage value, in particular, higher than the read voltage value. In other words, when the read voltage is applied to the gate 28, the programmed ferroelectric storage transistor is not conducting.

(49) To read the polarization state from the ferroelectric storage transistor, a small positive gate-to-source voltage (“the read voltage”) is applied and a small positive bit line voltage is applied. For example, the word line or gate voltage can be set to a positive voltage of 0.7V, the source line voltage set to 0V, and the bit line voltage set to 0.5V. As thus biased, a ferroelectric storage transistor that is in the erased state will be conducting and an “on” current I_{on} flows between the bit line and the source line of the erased storage transistor. Meanwhile, a ferroelectric storage transistor that is in the programmed state will be non-conducting and no current flows between the

bit line and the source line of the programmed storage transistor, except for a small leakage current, referred to as an “off” current I_{off} . To prevent disturb to the unselected ferroelectric storage transistors during the read operation, the unselected storage transistors have their gate voltage, source line voltage and bit line voltage all set to 0V. As thus biased, ferroelectric storage transistors selected for read operation that are erased will conduct current I_{on} while ferroelectric storage transistors selected for read operation that are programmed will conduct current I_{off} . The bit lines of the ferroelectric storage transistors are connected to sense amplifiers to sense the cell current and to generate an output logical value indicative of the cell current.

(50) In the above description, voltage bias conditions are described and example voltage values are provided to illustrate the erase, program and read operations of the memory array of ferroelectric storage transistors. The voltages described above are illustrative only and not intended to be limiting. In other embodiments, other voltage values can be used to implement the voltage bias conditions for erase, program and read operations.

(51) As described above, in embodiments of the present disclosure, the common source line of the memory structure can be left floating, or not electrically connected to any potential. Instead, the common source line can be pre-charged using one or both of the bit lines in the pair of NOR memory strings prior to erase, program and read operations. In one embodiment, the pre-charge operation set the bit line to a desired voltage through a pre-charge transistor and then the source line is charged from the bit line to a voltage to equal to the bit line voltage.

(52) Fabrication Process

(53) FIGS. 6(a) to 6(q) illustrate a process for fabricating a memory structure including a 3-dimensional array of NOR memory strings of ferroelectric storage transistors according to embodiments of the present invention. FIGS. 6(a) to 6(l) illustrate vertical cross-sectional views (i.e., in an X-Z plane) of a portion of the 3-dimensional array of NOR memory strings. Each figure in FIGS. 6(m) to 6(q) includes two views: view (i) is a horizontal cross-sectional view (i.e., in an X-Y plane) along line A-A' in view (ii), and view (ii) is a vertical cross-sectional view (i.e., in an X-Z plane) along line A-A' in view (i).

(54) Referring to FIG. 6(a), initially, a memory structure **100** is formed by successive depositions of a multilayer structure including (i) an isolation dielectric layer **105**, (ii) a first sacrificial layer **106**, (iii) an isolation dielectric layer **107**, and (iv) a second sacrificial layer **108** on the planar surface of a semiconductor substrate **102**. In some embodiments, a buffer oxide layer may be provided on the surface of the semiconductor substrate **102** before the isolation dielectric layer **105** and the sacrificial layer **106** are formed on the substrate **102**. In the present embodiment, the isolation dielectric layer **105** is silicon oxycarbon (SiOC) layer and may have a thickness of 30 nm. The first and second sacrificial layers **106** and **108** are each a silicon nitride layer and may have a thickness of 40 nm. The sacrificial layers **106**, **108** will be replaced by a conductive layer in metal replacement process in subsequent processing. The isolation dielectric layer **107** is a silicon oxide (SiO₂) layer and may have a thickness of 50 nm. FIG. 6(a) shows the memory structure **100** after the depositions of the initial layers of thin films. In the present illustration, eight multilayer structures are deposited. As described above, dimensions are provided in the present description for illustrative purposes and are not intended to be limiting. In actual implementation, any suitable thicknesses or dimensions may be used. The drawings are not necessarily to scale.

(55) Referring to FIG. 6(b), a first set of trenches **110** are formed in memory structure **100** using, for example, an anisotropic etch after a photo-lithographical patterning step. In some examples, the trenches **110** may be 70 nm wide, with a spacing of 190 nm apart. Referring to FIG. 6(c), with the first set of trenches exposing sidewalls of the active layers **111**, recesses **112** are formed in each active layer **111** through the trench openings. That is, the first and second sacrificial layers **106**, **108** and the isolation dielectric layer **107** are etched or recessed, such as by 10 nm, from the sidewall of the trenches **110**. A channel region is formed by depositing an amorphous metal oxide semiconductor layer **115** conformally on the exposed sidewalls of the trenches **110**, as shown in

FIG. 6(d). The amorphous metal oxide semiconductor layer **115** (“oxide semiconductor layer **115**”) maybe deposited using, for example, an atomic layer deposition (ALD) technique, a plasma-enhanced ALD technique or using physical vapor deposition (PVD). In particular, the oxide semiconductor layer **115** can be formed at a low processing temperature, such as lower than 400° C. In the present embodiment, the oxide semiconductor layer **115** is an indium gallium zinc oxide (InGaZnO, or “IGZO”) layer. For example, IGZO deposition using thermal ALD or plasma enhanced ALD can be carried out at a processing temperature as low as 200° C. Other oxide semiconductor materials that may be used include indium zinc oxide (InZnO, or “IZO”), indium aluminum zinc oxide (IAIZnO) or indium tin zinc oxide (ITZO). In one embodiment, the oxide semiconductor layer **115** has a thickness of 10 nm. The oxide semiconductor layer **115** will become the channel region of the ferroelectric storage transistors to be formed.

(56) Referring to FIG. 6(e), after deposition of the conformal oxide semiconductor layer **115**, the oxide semiconductor layer **115** is etched to separate the channel regions to each active layer **111**. For example, the oxide semiconductor layer **145** may be etched using a plasma dry etch process. As a result, the oxide semiconductor layer **115** forms channel regions on the sidewall of each active layer **111**, isolated or separated by the insulating dielectric layer **105**. Thereafter, the trenches are filled by a dielectric material **118**, as shown in FIG. 6(f). In some embodiments, the dielectric material **118** is a silicon oxide material, such as SiO₂. Excess dielectric material may be removed from the top of memory structure **100** using, for example, chemical-mechanical polishing (CMP).

(57) Referring to FIG. 6(g), with the mechanical support from dielectric material **118**, a second set of trenches **120** are then cut using substantially the same technique as discussed in conjunction with FIG. 6(b) above. In some examples, the trenches **120** may be 70 nm wide. Each of the second set of trenches **120** is cut between an adjacent pair of the first set of trenches **110** and each of the second set of trenches **120** is cut substantially equidistant between an adjacent pair of the first set of trenches **110**. As a result of trenches **110** and **120** being cut in the multilayer structure, stacks in the multilayer structure are formed which are referred to as “active stacks” in the present description. In some examples, the active stacks are each approximately 60 nm wide. The resulting narrow strips of layers **105** to **108** in the active stacks are referred herein as “active strips.”

(58) Thereafter, the first and second sacrificial silicon nitride layers **106** and **108** are removed using, for example, a wet etch, thereby creating cavities between the isolation dielectric layers **105** and **107**, as shown in FIG. 6(h). The cavities are filled by conductive layers **126**, **128**, as shown in FIG. 6(i). In some embodiments, the conductive layers **126**, **128** is a refractory metal layer, such as molybdenum (Mo) or tungsten (W). In other embodiments, the conductive layers **126**, **128** can be a metal layer selected from molybdenum, titanium, tungsten, lanthanum, tantalum, ruthenium, any silicide thereof, any nitride thereof, and any combination thereof. For example, the conductive layers **126**, **128** may be formed using, for example, an atomic layer deposition (ALD) technique. An etching step removes any of the deposited material from the sidewalls of trenches **120**. In some examples, an isotropic wet etch is used to remove the deposited material from the sidewalls of trenches **120**. In the present embodiment, the conductive layers **126**, **128** is a molybdenum layer and forms the source and drain terminals of the ferroelectric storage transistors to be formed.

(59) Thereafter, the exposed sidewalls of the active layers are etched again to form recesses **122**, such as by 10 nm, from the sidewall of the trenches **120**, as shown in FIG. 6(j). An amorphous metal oxide semiconductor layer **115** is again conformally deposited on the exposed sidewalls of the trenches **120**, such as using an ALD process. The oxide semiconductor layer **115** may have a thickness of 10 nm. The oxide semiconductor layer **115** are etched to separate the oxide semiconductor layer **115** to each active layer **111**, as shown in FIG. 6(k). Trenches **120** are then filled by the dielectric material **118**, such as SiO₂. Excess deposited materials may be removed from the top of memory structure **100**, such as by CMP. The resulting memory structure **100** is shown in FIG. 6(l).

(60) Referring to FIG. 6(m), thereafter, vias **124** are patterned and the dielectric material **118** exposed by the vias **124** are etched using, for example, an anisotropic etch. Vias **124** may be masked in an oval shape, as shown in the horizontal cross-sectional view (i) of FIG. 6(m). In some examples, vias **124** may have a major axis of 100 nm along the X-direction and a minor axis of 60 nm along the Y-direction. Vias **124** are filled at a later time by a dielectric material so as to isolate adjacent storage transistors in an NOR memory strings to be formed. In the present embodiment, the etch process to form vias **124** also remove the oxide semiconductor layer **115** from the vias in addition to the dielectric material **118**. In this manner, the oxide semiconductor layer **115** is separated in the Y-direction along the NOR memory string, as shown in view (i) of FIG. 6(m).

(61) Referring to FIG. 6(n), vias **124** are then filled with a sacrificial silicon nitride liner and a sacrificial amorphous silicon layer, collectively referred to as layer **130**. Excess materials on top of memory structure **100** may be removed using, for example, CMP. With the vias **124** filled, the remainder of dielectric material **118** in trenches **110** and **120** is then removed using, for example, a wet etch step. As a result, cavities are formed in trenches **110** and **120** outside of the vias **124** which are now filled with sacrificial layers of silicon nitride liner and amorphous silicon layer **130**.

(62) Referring to FIG. 6(o), a ferroelectric dielectric layer **136** is then conformally deposited on the exposed sidewalls of the excavated cavities between the filled vias **130**. In one embodiment, the ferroelectric dielectric layer **136** is deposited directly onto the oxide semiconductor layer **115**. In some embodiments, the ferroelectric dielectric layer **136** is deposited by any one of suitable deposition methods including but not limited to chemical vapor deposition (CVD), low pressure chemical vapor deposition (LPCVD), physical vapor deposition (PVD) and atomic layer deposition (ALD). Other deposition methods such as evaporation and electrodeposition can also be used. In some embodiments, the ferroelectric dielectric layer **136** is a doped hafnium oxide (HfO₂) layer. In one embodiment, the hafnium oxide is doped with zirconium oxide (ZrO₂) to form a hafnium zirconium oxide layer (HfZrO or “HZO”). In another embodiment, the hafnium oxide is doped with silicon (Si), iridium (Ir) and lanthanum (La). In some embodiments, the ferroelectric dielectric layer **136** is a material selected from: zirconium-doped hafnium oxide (HZO), silicon-doped hafnium oxide (HSO), aluminum zirconium-doped hafnium oxide (HfZrAlO), aluminum-doped hafnium oxide (HfO₂:Al), lanthanum-doped hafnium oxide (HfO₂:La), hafnium zirconium oxynitride (HfZrON), hafnium zirconium aluminum oxide (HfZrAlO), and any hafnium oxide that includes zirconium impurities.

(63) In one embodiment, the ferroelectric dielectric layer **136** has a thickness of 4 nm. In some embodiments, an interface dielectric layer is formed between the oxide semiconductor layer **115** and the ferroelectric dielectric layer **136**. In some embodiments, the interface dielectric layer is formed using a material with a high permittivity (K) (also referred to as “high-K” material). In one example, the interface dielectric layer, if present, may be a silicon nitride (Si₃N₄) layer and has a thickness of 1 nm. The ferroelectric dielectric layer **136** may be deposited and then annealed to form the ferroelectric phase in the ferroelectric material. The ferroelectric dielectric layer **136** forms the gate dielectric layer of the ferroelectric storage transistors to be formed.

(64) After the deposited ferroelectric dielectric layer **136** is annealed, the cavities between the filled vias **130** are filled by a conductive layer **138**, which may include successively deposited titanium nitride (TiN) liner and a tungsten (W) layer. The TiN liner may be formed using, for example, an atomic layer deposition (ALD) technique. In some embodiments, the annealing of the ferroelectric dielectric layer **136** is performed after the conductive layer **138** is deposited. As thus formed, the conductive layer **138** provides a vertical local word line (LWL) that serves as gate electrode for each of the ferroelectric storage transistors that are vertically aligned in the same active stack. The structure resulting from the combination of the ferroelectric dielectric layer **136** and the conductive layer **138** is referred herein as the local word line (LWL) structure. Excess deposited materials may be removed from the top of memory structure **100**, such as by CMP. FIG. 6(o) illustrates the resulting memory structure **100**.

(65) It is instructive to note that the memory structure **100** includes regular sized cavities in a central portion of the memory structure and extended sized cavities at the edges of the memory structure. As a result, the memory structure **100** includes LWL structures formed in the regular sized cavities which are to be used to form the ferroelectric storage transistors of the NOR memory strings. The structures formed in the extended sized cavities at the edges of the memory structure **100** can be dummy structures or they can be converted to non-memory transistors, where applicable. The exact configuration of the memory structure **100** at the edges of the memory strings is not critical to the practice of the present invention.

(66) Referring to FIG. **6(p)**, subsequent to the LWL structures being formed, the sacrificial materials formed in the vias **124** are now removed, forming cavities **140** between the LWL structures. In the present embodiment, the amorphous silicon layer is removed, such as by using a wet etch step. The silicon nitride liner is left in the cavities **140**. In other embodiments, the silicon nitride liner may also be removed, such as by using a wet etch step.

(67) Referring to FIG. **6(q)**, a dielectric material **150** then fills the exposed vias **140**. In some embodiments, the dielectric material **150** is silicon oxide, such as SiO₂. Excess material on top of memory structure **100** may be removed by CMP. The resulting memory structure **100** includes multiple layers of NOR memory strings of ferroelectric storage transistors formed in multiple active stacks to realize a memory device with large capacity. In particular, the ferroelectric storage transistors in the NOR memory strings are formed by the conductive layer **126** as the source terminal, the conductive layer **128** as the drain terminal, the oxide semiconductor layer **115** as the channel region, the ferroelectric dielectric layer **136** as the gate dielectric layer and the conductive layer **138** as the gate terminal or gate conductor.

(68) In this detailed description, process steps described for one embodiment may be used in a different embodiment, even if the process steps are not expressly described in the different embodiment. When reference is made herein to a method including two or more defined steps, the defined steps can be carried out in any order or simultaneously, except where the context dictates or specific instruction otherwise are provided herein. Further, unless the context dictates or express instructions otherwise are provided, the method can also include one or more other steps carried out before any of the defined steps, between two of the defined steps, or after all the defined steps.

(69) In this detailed description, various embodiments or examples of the present invention may be implemented in numerous ways, including as a process; an apparatus; a system; and a composition of matter. A detailed description of one or more embodiments of the invention is provided above along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. Numerous modifications and variations within the scope of the present invention are possible. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications, and equivalents. Numerous specific details are set forth in the description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured. The present invention is defined by the appended claims.

Claims

1. A three dimensional memory structure formed above a planar surface of a semiconductor substrate, the memory structure comprising: a plurality of thin-film ferroelectric field-effect transistors (FeFETs) formed above the semiconductor substrate and arranged in one or more parallel planes, wherein (i) the thin-film FeFETs are organized as NOR memory strings, each NOR memory string extending along a first direction substantially parallel to the planar surface of the

semiconductor substrate; (ii) the NOR memory strings are arranged in two directions: (a) as stacks of NOR memory strings, each stack including NOR memory strings arranged one on top of another along a second direction substantially normal to the planar surface, the NOR memory strings within each stack being isolated from another by a first isolation layer separating adjacent planes, and (b) as rows of NOR memory strings arranged along a third dimension substantially orthogonal to both the first dimension and the second direction; (iii) the FeFETs within each NOR memory string share a common source line and a common bit line, the common source line and the common bit line each extending along the first direction and arranged spaced apart from each other in the second direction by a second isolation layer; (iv) each FeFET in each NOR memory string includes an oxide semiconductor layer and a ferroelectric gate dielectric layer formed adjacent the oxide semiconductor layer, the oxide semiconductor layer forming a channel region between the common source line and the common bit line, the oxide semiconductor layer being separated by the first isolation layer to each NOR memory string within each stack; and (v) a plurality of conductors each extends along the second direction, each conductor being provided between adjacent stacks of NOR memory strings and serves as a common gate electrode for respective FeFETs in the NOR memory strings of the adjacent stacks.

2. The memory structure of claim 1, wherein the ferroelectric gate dielectric layer comprises a ferroelectric polarization layer provided adjacent each conductor as a continuous layer in the second direction.

3. The memory structure of claim 1, wherein the ferroelectric gate dielectric layer comprises a ferroelectric polarization layer provided adjacent each conductor, the ferroelectric polarization layer associated with each FeFET in each stack of NOR memory strings being separated from the ferroelectric polarization layer of other FeFETs along the stack in the second direction.

4. The memory structure of claim 1, wherein the oxide semiconductor layer comprises one of an indium zinc oxide (IZO) layer and an indium gallium zinc oxide (IGZO) layer.

5. The memory structure of claim 4, wherein the ferroelectric gate dielectric layer contacts the oxide semiconductor layer directly without an interface dielectric layer therebetween.

6. The memory structure of claim 1, wherein the oxide semiconductor layer and the ferroelectric gate dielectric layer are formed adjacent a respective conductor between adjacent stacks and the oxide semiconductor layer is isolated in the first direction from the oxide semiconductor layer associated with other conductors provided between the same adjacent stacks.

7. The memory structure of claim 1, wherein the common source line and the common bit line each comprises a metal layer.

8. The memory structure of claim 7, wherein the oxide semiconductor layer associated with the FeFETs in each NOR memory string contacts the metal layers forming the common source line and the common bit line to provide a junctionless channel region for each FeFET.

9. The memory structure of claim 7, wherein the metal layer comprises the metal comprises one or more of: molybdenum, titanium, tungsten, lanthanum, tantalum, ruthenium, any silicide thereof, any nitride thereof, and any combination thereof.

10. The memory structure of claim 1, wherein the plurality of conductors each comprises a titanium nitride layer.

11. The memory structure of claim 10, wherein the plurality of conductors each comprises the titanium nitride layer formed adjacent the ferroelectric gate dielectric layer and a tungsten layer formed adjacent the titanium nitride layer.

12. The memory structure of claim 1, wherein the ferroelectric gate dielectric layer comprises a doped hafnium oxide layer.

13. The memory structure of claim 12, wherein the doped hafnium oxide layer comprises one or more of: zirconium-doped hafnium oxide (HZO), silicon-doped hafnium oxide (HSO), aluminum zirconium-doped Hafnium oxide (HfZrAlO), aluminum-doped hafnium oxide (HfO.sub.2:Al), lanthanum-doped hafnium oxide (HfO.sub.2:La), hafnium zirconium oxynitride (HfZrON),

hafnium zirconium aluminum oxide (HfZrAlO) and any hafnium oxide that includes zirconium impurities.

14. The memory structure of claim 1, wherein each FeFET is electrically placed in a first polarization state by applying a first voltage to both the common source line and the common bit line and a second voltage to the associated common gate electrode; and each FeFET is electrically placed in a second polarization state by applying a third voltage to the common bit line, a fourth voltage to the common source line and a fifth voltage to the associated common gate electrode, the third voltage being different from the fourth voltage.

15. The memory structure of claim 1, wherein each FeFET is electrically placed in a first polarization state by applying a first voltage to both the common source line and the common bit line and a second voltage to the associated common gate electrode; and each FeFET is electrically placed in two or more other polarization states by applying voltages to the common bit line, the common source line and the associated common gate electrode, the voltages having a different voltage range for each of the two or more other polarization states.

16. The memory structure of claim 14, wherein the common source lines of the NOR memory strings are electrically floating and a respective common source line is biased to a given voltage during a pre-charge phase and is left floating afterwards.

17. The memory structure of claim 1, wherein each stack of NOR memory strings comprises FeFETs formed on the sidewalls on both sides of the stack.

18. The memory structure of claim 1, wherein circuitry for supporting memory operations is formed at the planar surface of the semiconductor substrate underneath the plurality of stacks of FeFETs.

19. The memory structure of claim 18, wherein the circuitry for supporting memory operations implements erase, program and read operations for the plurality of stacks of FeFETs in response to erase, program and read commands provided to the memory structure.

20. The memory structure of claim 18, further comprising a layer of interconnect conductors formed above and in electrically connection with the NOR memory strings and the circuitry for supporting memory operations, the layer of interconnect conductors being provided for routing control and data signals among the NOR memory strings and the circuitry for supporting memory operations.

21. The memory structure of claim 20, wherein the circuitry for supporting memory operations comprises two or more of: word line driver circuits, bit line driver circuits, input/output driver circuits, address decoders, sense amplifiers, voltage sources for generating operating voltages for the memory operation, latches, registers, or other memory elements, and a state machine for managing the memory operations at the FeFETs of the NOR memory strings.

22. The memory structure of claim 21, wherein the circuitry for supporting memory operations is coupled to a controller integrated circuit formed on a separate semiconductor substrate to receive the command and data signals from the controller integrated circuit.
